



Chapter 19 — Starter Content, Licensing & Zero-Cost Stack



This chapter closes the remaining gaps: the starter scenario library, the AI prompt library, licensing & legal decisions, the full project cost sheet, and a category-by-category audit proving the entire platform can be built and run on **free and open-source tools**.

19.1 Starter Scenario Library (ships with Phase 1–3)

The engine is only as good as its scenarios. These eight ship in `scenarios/` as YAML — ordered by build phase:

#	Scenario	Storyline	Entities	Duration (sim)	Showcases
SC-1	Night Patrol Baseline	Quiet sector patrol; 2 minor incidents	12	2 h	Phase 1 smoke test: movement, feed, map
SC-2	Border Surveillance & Infiltration Response	Sensor line trips along a border sector; suspected infiltration; QRF dispatch and cordon	35	6 h	<i>Flagship demo.</i> Zones, severity escalation, sensor fusion, 5-km ranking
SC-3	Counter-UAV Airspace Watch	Unidentified drone incursions over restricted airspace; detection review and intercept tasking	28	4 h	Missions, CV pipeline, dynamic restricted zones
SC-4	Convoy Escort Under Threat	Supply convoy re-routing around ambush-risk zones and worsening storm cells	20	5 h	A* re-planning, threat/weather cost layers, feasibility

#	Scenario	Storyline	Entities	Duration (sim)	Showcases
SC-5	Forward Post Casualty Evacuation	Engagement produces casualties across two posts; medevac unit saturation	40	3 h	Task assignment, logistics burn-down, vitals
SC-6	Extended Logistics Sustainment	Forward positions consume fuel/medical faster than resupply cycles	30	12 h	Forecasting, low-stock alerts, analytics
SC-7	Multi-Sector Escalation Exercise	Coordinated incidents across 4 border sectors simultaneously; not enough units	60	8 h	War-room mode, sector ops, RTA pressure, NFR-2 load
SC-8	HADR Deployment — Flood Relief	Military aid-to-civil-authority flood operation: evacuation, medical, engineering tasks	25	6 h	Dynamic geometry, replay & What-If, Ch. 14 benchmarking link

Each scenario file declares: metadata (name, description, difficulty), map bounds & sectors, entity roster, scripted timeline events, win/end conditions, and default seed. SC-2 is the canonical golden-run fixture (Ch. 11). SC-8 is kept deliberately: HADR is a standing military mission worldwide, and it is the only scenario class with open real-world outcome data — which Chapter 14's benchmarking requires.

19.2 RAG Starter Corpus (SOPs & references)

Written as original mock documents (no copyright issues), stored in `corpus/`, ~15 documents:

- SOP-01..08: patrol & contact reporting, QRF dispatch, casualty evacuation (medevac), convoy escort procedures, counter-UAS response, comms-loss protocol, base perimeter defense, HADR deployment
- REF-01..04: unit type capabilities sheet, severity classification guide, sector map legend, escalation matrix
- POL-01..03: rules of engagement (simulated), data handling policy, after-action report template

Each doc is 1–3 pages, headed with metadata (id, title, tags, version) for chunking — realistic enough for convincing RAG demos, small enough to author in two evenings.

19.3 Prompt Library

Prompts are **versioned artifacts**, not scattered strings: they live in `ai-service/prompts/*.md` with YAML frontmatter (id, model, version, eval score) and are loaded at runtime. The canonical set:

Prompt	Pipeline	Key constraints baked in
<code>sitrep-author</code>	Situation reports	Only cite provided events; every claim tagged <code>[e:<id>]</code> ; ≤250 words; severity-ordered
<code>grounding-verifier</code>	Sitrep QA gate	Cross-check each claim against event list; flag unsupported claims
<code>rag-answerer</code>	Operator Q&A	Answer only from retrieved chunks; cite doc ids; say “not in SOPs” when absent
<code>query-router</code>	Q&A entry	Classify: live-state query → SQL path; doctrine query → RAG path; hybrid → both
<code>scenario-author</code>	What-If Theater (Ch. 13)	Emit only whitelisted DSL actions as JSON; no free-text effects
<code>playback-narrator</code>	Animated replay (Ch. 13)	Narrate provided event windows; past tense; no speculation
<code>mission-brief</code>	Mission summaries	Structured brief: objective, route, risks, fallback
<code>rta-flavor</code> (v2, optional)	Red-team injects (Ch. 16)	Flavor text only — mechanics come from the playbook, never the LLM

Example skeleton (`sitrep-author`):

```
You are the DOIP situation report author.
INPUT: a list of events, each with id, sim-time, type, severity, location.
RULES:
1. Use ONLY the provided events. Do not infer unstated facts.
2. Tag every claim with its source: [e:1042].
3. Order by severity, then recency. Max 250 words.
4. End with "ASSESSMENT:" — one sentence, clearly marked as inference.
OUTPUT: markdown with sections CRITICAL / ONGOING / RESOLVED / ASSESSMENT.
```

Prompt changes require an eval re-run (Ch. 11 grounding suite) — same discipline as code.

19.4 Licensing & Legal

Decision	Choice	Rationale
Project license	Apache-2.0	Permissive like MIT but adds an explicit patent grant; the standard for infra/AI projects
Ultralytics YOLOv8/v11	 AGPL-3.0	Fine only if the project stays open-source (it does). If a closed fork is ever wanted: swap to YOLOX (Apache-2.0) or RT-DETR via PaddleDetection (Apache-2.0) — the Vision Service's model interface (Ch. 8) makes this a contained change

Decision	Choice	Rationale
Llama 3 8B	Meta community license	Free for this use; not OSI-open — documented, with Qwen as alternative
Qwen 2.5 7B	Apache-2.0	Cleanest LLM option; default for redistribution safety
Map tiles	OpenFreeMap (free) / OSM	ODbL attribution required: “© OpenStreetMap contributors” rendered on every map view — non-negotiable
MinIO	AGPL-3.0	Used as unmodified server over S3 API — no copyleft impact on DOIP code; SeaweedFS (Apache-2.0) listed as fallback
VisDrone dataset	Academic/research use	Used for fine-tuning demos only; noted in README; synthetic frames are the default demo path
Fonts (Inter, JetBrains Mono)	SIL OFL	Free to bundle
Disclaimer	In README + login screen	“Training & simulation platform. All data is synthetic. Not for operational military use.”

Repo additions: `LICENSE` (Apache-2.0), `NOTICE` (attributions), `THIRD_PARTY_LICENSES.md` (generated via `pip-licenses` / `license-checker` — both free).

19.5 Project Cost Sheet

Software: ₹0. Every runtime component, library, model, and tool in Chapters 4 and 18 is free. Complete honest cost picture:

Item	Mandatory?	Cost
All software, models, tools	—	₹0
Dev machine (existing laptop, 8–16 GB)	Yes	₹0 (already owned)
Electricity (~heavy dev use)	Yes	~₹200–400/month
GitHub (public repo: Actions 2,000 min/mo, GHCR, Pages)	Yes	₹0
Public demo hosting — Option A: own laptop + Cloudflare Tunnel (free) during demos	No	₹0
Public demo hosting — Option B: Oracle Cloud Always-Free ARM VM (4 OCPU/24 GB)	No	₹0 (best-effort availability)
Public demo hosting — Option C: budget VPS	No	~₹400–1,500/month
Domain name	No	~₹800–1,000/year
Hosted LLM API for premium demo answers	No	₹0 default (Ollama); pay-per-use only if ever chosen

Total to build everything: ₹0 beyond electricity. The optional line items exist only for a public 24/7 demo, and even that has two ₹0 paths.

19.6 Zero-Cost Stack Audit (category by category)

Category	Tools	License	Free?
Frontend	React, Vite, MapLibre, Zustand, TanStack, Tailwind, shadcn/ui	MIT/BSD	✓
Map data	OpenStreetMap, OpenFreeMap tiles	ODbL (attribution)	✓
Backend	Python, FastAPI, Pydantic, SQLAlchemy, Celery	MIT/BSD/Apache	✓
Data	PostgreSQL, PostGIS, pgvector, Redis, Kafka, MinIO	Open source	✓
AI/ML	Ollama, Qwen/Llama, LangGraph, sentence-transformers, XGBoost, Prophet, MLflow	Apache/MIT/community	✓
CV	YOLO (AGPL — see §19.4), OpenCV	AGPL/Apache	✓
DevOps	Docker, Compose, k3s, GitHub Actions, Nginx, Prometheus, Grafana, Loki	Open source / free tier	✓
Design	Excalidraw, Penpot, Lucide, Inter/JetBrains Mono	MIT/MPL/ISC/OFL	✓
Testing	pytest, Playwright, k6 (load)	MIT/Apache/AGPL	✓
Docs & PM	This Notion doc, GitHub issues/projects, Markdown in repo	Free tiers	✓

Standing rule for all future additions: any new dependency must be (a) free for this use, (b) license-compatible with Apache-2.0 distribution (or isolated behind an interface like YOLO), and (c) recorded in `THIRD_PARTY_LICENSES.md` — checked in CI.

 **Starter Artifact Pack — Scenarios, SOP Corpus & Footage**